Specify the Business Problem

Business Problem

Agriculture is one of the most important sectors of the economy, yet many farmers still rely on traditional farming practices and limited access to scientific decision-support systems. Factors such as soil nutrient imbalance, unpredictable weather conditions, improper irrigation, pest attacks, and lack of timely expert guidance often result in low crop productivity and financial losses.

Small and medium-scale farmers frequently struggle to determine the most suitable crop for their land because they lack access to accurate agricultural information and modern decision-support systems. Existing agricultural advisory services are often expensive, limited in accessibility, or unavailable in local languages.

The objective of this project is to develop an intelligent Smart Agricultural Production System that leverages Machine Learning to assist farmers in selecting suitable crops. The system analyzes soil nutrients (Nitrogen, Phosphorus, Potassium), pH, temperature, humidity, and rainfall to recommend the most appropriate crop for cultivation based on the trained Machine Learning model.

The proposed system aims to improve agricultural productivity, reduce crop failure, optimize resource utilization, and promote sustainable farming practices.